DPM Atomic Creation Process (ACP): Proto-Nucleus Formation

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PAPER_738 — DPM Atomic Creation Process (ACP): Proto-Nucleus Formation

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Title: Di-Pseudo-Monopole (DPM) Creation Scenario and the Atomic Creation Process (ACP): From DPM Initiation to Proto-Nucleus Assembly
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Source: thread_06Jun2025.txt (lines 8100–8387, June 06, 2025, "describe mass without using weight.docx" + "05June2025.docx")

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1. Abstract

The Atomic Creation Process (ACP) establishes the origin of fundamental particles through Di-Pseudo-Monopole (DPM) mediated reactions, replacing the Standard Model's mass-as-weight construct with a buoyancy-based framework. The ACP is a six-stage process beginning with DPM formation and culminating in proto-nucleus assembly with 26 quantum states, mediated by Universal Aether (UA') and Super Conductive Material (SCm). Atomic mass does not emerge until the quantum-to-mass gradient at 7–10 U_mag degrees.

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2. Three Fundamentals of UQFF

The entire universe is built from three elemental interactions:

1. **Electrostatic Barrier (REB)** — The boundary condition that prevents collapse, defining system shells
2. **Universal Aether (UA')** — Non-local connectivity medium, vacuum energy carrier, ρ_UA' = 7.09e-36 J/m3
3. **Super Conductive Material (SCm)** — Superconductive magnetism mediator, ρ_SCm = 7.09e-37 J/m3

$$DPM = UA' + SCm$$

The Di-Pseudo-Monopole (DPM) is formed by the interaction of UA' and SCm, producing vacuum density at any scale from atomic to cosmic.

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3. The Six-Stage Atomic Creation Process (ACP)

Stage 1: DPM Initiation

$$UA' + SCm \rightarrow DPM(Di - Pseudo - Monopole)$$
$$DPM \rightarrow vacuumdensity\rho_vac$$

- UA' provides non-local field connectivity - SCm provides superconductive coherence - Product: localized vacuum energy density (capacitance) → proto-nuclear seed - Billions of coherent DPM nuclear cells operate simultaneously at every location

Stage 2: U_i Creation (Intelligent Operator)

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DPM \rightarrow U_i  (repulsive + intelligent quantum operator)
```

- U_i is the first active particle: repulsive (buoyant), self-directing
- U_i orchestrates all subsequent proto-nuclear assembly
- Creates the fundamental "anti-gravity" seed that all matter requires to exist

Stage 3: U_m String Formation

$$U_i \rightarrow U_m strings(\text{U\_mag\_i} = magneticstringsequences)$$

- U_i creates coherent U_m (U_{mag_i}) strings - Strings wind concentrically around the vacuum density core → proto-nucleus - String winding number corresponds to quantum state number (1..26) - Each winding creates one of the 26 quantum states (Russian doll nesting)

Stage 4: Proto-Shell Formation and Cracking

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Vacuum density grows \rightarrow vacuum energy density (capacitance threshold)
\rightarrow quantum ripples:  ULF_quantum^{-1,...,-26}
\rightarrow proto-shell cracks into fragments
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- When vacuum energy density reaches critical capacitance:
  - Quantum ripples: ULF_quantum^{-1} through ULF_quantum^{-26}
  - Proto-shell fractures into 26 distinct fragment types
  - Fragments: proto-quarks, proto-bosons, proto-leptons
- This is not "particle creation from nothing" — it is structured fracturing of a pre-existing quantum state container

Stage 5: Fragment Stabilization

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SM_magnetic moments (SM_mag) \rightarrow arrange fragments on durable nucleus
U_b (buoyancy) maintains stability against SM_gravity
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- SM_magnetic moments act as “sorting fingers” that arrange proto-fragments
- U_b (Universal Buoyancy, massless) continuously counteracts SM_gravity
- Result: stable proto-nucleus with 26-state organization
- Still pre-mass at this stage

Stage 6: Electron Shell Formation → Mass Emergence

$$U_g 2 governs electron shell placement via U_m$$
$$Mass emerges at quantum - to - mass gradient : 7 - -10 U_m ag degrees$$
$$H(1) = first hydrogen atom formed$$

- U_g2 uses U_m string frequency to place electrons into shells - Atomic mass is NOT weight; it is the ratio of effective gravity to superconductive buoyancy in Earth’s field - Mass emerges when the quantum state gradient crosses 7–10 U_mag degrees of superconductive magnetism

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4. Mass Without Weight — Formal Definition

$$Mass(UQFF) = proportional to :$$
$$(\text{SM\_effective\_gravity\_range})/(superconductive_b uoyancy)$$
$$= FU_g 1/\text{F\_U\_Bi}[ratio of gravitational to buoyant components]$$

Examples of mass-as-buoyancy: - Bi-molecule: hydrogen bonds mediated by DPM electrostatic barrier - Earth-Moon: orbital buoyancy balanced by tidal resonance shells - Sun-SagA*: Galactic-scale DPM coherence, U_g4i THz hole connection - Universal scale: all bodies are buoyant in the Aether medium

Mass is never a stand-alone measurement. It is only meaningful as the ratio of: - The Effective Gravity force (SM_gravity, FU_g1) - To the Effective Buoyancy (U_b, F_{U_Bi})

$$"Mass" = FU_g 1/\text{F\_U\_Bi}(dimensionless ratio, context - dependent)$$

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5. U_b Tracking → f_Ub Definition

During ACP, U_b is tracked through calibration differences:

$$f_{Ub} \propto \Delta k_\eta \quad \text{(deviation from expected calibration constant } k_\eta)$$
$$k_{\eta,\text{expected}} \approx 10^9 \quad \text{(galaxy-scale)}$$
$$k_{\eta,\text{actual}} = \text{measured value from observation}$$
$$\Delta k_\eta = k_{\eta,\text{expected}} - k_{\eta,\text{actual}}$$
$$f_{Ub} = \Delta k_\eta / k_{\eta,\text{reference}}$$

Scale	f_Ub order
Galaxy	~10^9

Scale	f_Ub order
Dwarf galaxy / LMC	~10^8
Star cluster	~10^7
PN / planetary	~10^5
Atomic	~10^3

The imaginary/quantum portion of gravity (U_b) is never zero — it is this hidden buoyancy that modern physics has failed to account for, leading to the need for "dark matter" and "dark energy" as placeholder terms. UQFF resolves both as manifestations of f_Ub at different scales.

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6. 26 Quantum States — Pre-Mass Configuration

The 26 quantum states are NOT the electron orbitals of quantum mechanics. They are:

$$
\begin{aligned}
&Statei(i = 1..26) : pre - massnuclearconfigurationlevel \\
&- Shellradius : r_i = r_system/i \\
&- Superconductivemagnetism : [SCm]_i = 1e - 5 * i2T \\
&- Angularcoverage : \theta_i = 90^\circ - (i - 1) * 3.346^\circ (spans90^\circ \rightarrow\ 5^\circ) \\
&- THzholeradius : r_THz, i = 1e - 9/im \\
&- Resonancefactor : f_TRZ, i = 0.1 * i/26 \\
&- Buoyancycoupling : f_Um, i = 0.05 * i/26
\end{aligned}
$$

Like Russian dolls, each state contains and surrounds the previous. State 26 is the outermost (most magnetic, widest angular coverage initially); State 1 is the innermost quantum seed.

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7. DPM Coherence — The Source of Predictability

$$
\begin{aligned}
&Coherencelength = "allatomsinabodysharethesameDPMphase" \\
&NumberofcoherentDPMspercubiccentimeter\ 10^23(Avogadro - scale) \\
&Result : macroscopicquantumphenomena \rightarrow superconductivityatscale
\end{aligned}
$$

This coherence is why: - Planetary rings form symmetric arcs - Stars form in filaments rather than randomly - Galaxies maintain spiral structure for billions of years - LENR occurs at measurable, reproducible rates

All of these are manifestations of coherent DPM nuclear cell behavior.

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8. ACP Mathematical Summary

$$Step1 : DPM = UA' + SCm$$
$$Step2 : U_i = k_\eta * (\rho_v ac, SCm - \rho_v ac, UA) * \omega_i * cos(\pi * t_n)$$
$$Step3 : U_m, i = U_i * \mu_d ipole * (1/r_i) * (1 - e^(- \gamma t)) * cos(\pi t_n)$$
$$Step4 : \Psi_p roto = \Sigma_{i=1}^{26} U_m, i \rightarrow capacitancethresholdreached$$
$$Step5 : F_s tab = SM_m ag * \Psi_p roto + U_b \rightarrow U_b = -\beta * Ug * \Omega_g * (M_b h/d_g) * cos(\pi t_n)$$
$$Step6 : E_s hell = c * \nu_r es * h(f_S Cm) * G_g eo \rightarrow atomicmassemerges$$

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9. References

- Source: thread_06Jun2025.txt (June 06 2025), "describe mass without using weight.docx", "05June2025.docx"
- Related: PAPER_735 (Ug2 electron shell energy), PAPER_734 (LENR K_n), CP3 DiPseudoMonopoleDPMTheoryCalculator
- CP4 class: #322 DPMAtomicCreationProcessACPCalculator
- CVW v2.0.0 compliant

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ J/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **cosmogenesis** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.`

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u\phi_{\text{cosmo}})(\partial^\mu \phi_{\text{cosmo}}) - V(\phi_{\text{cosmo}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{cosmo}}) = \frac{1}{2}m^2\phi_{\text{cosmo}}^2 + \frac{\lambda}{4!}\phi_{\text{cosmo}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{cosmo}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{cosmo}}} = \partial_t(\rho_{\text{vac}} V_{\text{proto}}) + U_i + \Psi_{\text{proto}}/26 = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\phi_{\text{cosmo}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.156 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 67, \quad n_{\mathrm{channel}} = 11/26$$

Since $p_{\mathrm{DVP}} = 67$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **ACP Stage 4 timescale** (capacitance cracking epoch):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.156	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\ldots,113\}$	$p_{\mathrm{DVP}} = 67$	PASS Resonant
BSH layers	26 harmonic terms	j = 1. . . 26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1×10-52 m-2 (UQFF vacuum term)	1.114×10-52 m-2	Planck 2018	PASS Consistent
Proton decay rate	κ = 0.0005/day → Γ__p suppression	< 4.17×10-35/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_{U\_Bi\_i}` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_{U_Bi_i}) that produce measurable deviations from GR at scales where vacuum condensate density ρ__SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

16 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER__840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron__{s26__coupling}.py	F__neutron x S__26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima__{scm__cross__section}.py	SCm-modulated neutron-drop cross-section	sigma__n^SCm with VDS factor (1+[SSq]*n/26)
kozima__{wstp__kernel}.py	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F__neutron^SCm = N__n * sigma__n^SCm(omega) * Phi__phonon * (F__{U,Bi}/F__U - 1) where sigma__n^SCm(omega,n) = sigma__0 * exp[-(omega-omega__SCm)$^{2/(2*\mathrm{Gamma}}$2)] * (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan__{polylog__s26}.py	Li__26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_{wstp\_kernel}.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: S__26(z) = Li__26(z) = eta__26(z)/(1-2^{1-26}) + 2$^{\{1\text{-}26\}/(1\text{-}2}${1-26}) * Li__26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock__{theta__q26}.py	f__26(q), phi__26(q), psi__26(q) q-series	Proper q-Pochhammer (a;q)__n

Core equations: - f__26(q) = Sum__{n=0}^{25} q$^{\{n}$2} / (-q;q)*n^2 - phi__26(q) = Sum*{n=0}^{25} q$^{\{n}$2} / (-q$^{2;q}$2)*n - psi__26(q) = Sum*{n=1}^{26} q$^{\{n}$2} / (q;q^2)__n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_{pi_uqff}.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_{theta_pi_wstp_kernel}.py`	0-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n * (1103+26390n) * W_26(n) / C_26 where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

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